

# Depth-First Search (DFS) Using Python

## Aim

To write a Python program to implement the **Depth-First Search (DFS)** algorithm for traversing a graph by visiting each node as deeply as possible before backtracking.

## Algorithm

1. Start the program.
2. Define the graph using an adjacency list.
3. Create an empty set to store the visited nodes.
4. Start the DFS traversal from the given source node.
5. Mark the current node as visited and display it.
6. Recursively visit each unvisited adjacent node.
7. Continue until all reachable nodes are visited.
8. Stop the program.

## Python Code

```
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
}

visited = set()

def dfs(node):

    if node not in visited:
        print(node, end=" ")
        visited.add(node)
        for n in graph[node]:
            dfs(n)

dfs('A')
```

EXP 8( DFS Algorithm).py > ...

```
1 graph = {
2     'A': ['B', 'C'],
3     'B': ['D', 'E'],
4     'C': ['F'],
5     'D': [],
6     'E': [],
7     'F': []
8 }
9 visited = set()
10 def dfs(node):
11     if node not in visited:
12         print(node, end=" ")
13         visited.add(node)
14         for n in graph[node]:
15             dfs(n)
16 dfs('A')
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PS E:\OPEN CV\AI Lab> & C:\Users\user\AppData\Local\Pro  
A B D E C F